Green Chemical Engineering 6 (2025) 275–287

Contents lists available at ScienceDirect

Green Chemical Engineering

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Article

Physics-informed machine learning to predict solvatochromic parameters of designer solvents with case studies in CO_2 and lignin dissolution

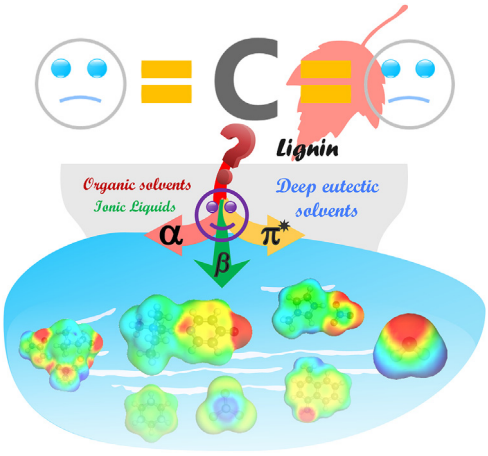
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HIGHLIGHTS

- Large and diverse dataset was compiled for Kamlet-Taft parameter predictions.
- Machine learning predicts Kamlet-Taft parameters for diverse designer solvents.
- FFNN model outperforms MLR with high R^2 and low RMSE in Kamlet-Taft parameter predictions.
- High basicity solvents show increased lignin and CO_2 solubility as predicted.
- SHAP analysis reveals HB acceptor moment as key for basicity prediction.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Kamlet-Taft parameter
Machine learning
Ionic liquids
Deep eutectic solvents
Organic compounds

ABSTRACT

The polarity of solvents plays a critical role in various research applications, particularly in their solubilities. Polarity is conveniently characterized by the Kamlet-Taft parameters that is, the hydrogen bonding acidity (α), the basicity (β), and the polarizability (π^*). Obtaining Kamlet-Taft parameters is very important for designer solvents, namely ionic liquids (ILs) and deep eutectic solvents (DESs). However, given the unlimited theoretical number of combinations of ionic pairs in ILs and hydrogen-bond donor/acceptor pairs in DESs, experimental determination of their Kamlet-Taft parameters is impractical. To address this, the present study developed two different machine learning (ML) algorithms to predict Kamlet-Taft parameters for designer solvents using quantum chemically derived input features. The ML models developed in the present study showed accurate predictions with high determination coefficient (R^2) and low root mean square error (RMSE) values. Further, in the context of present interest in the circular bioeconomy, the relationship between the basicities and acidities of designer solvents and

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https://doi.org/10.1016/j.gce.2024.11.003
Received 3 September 2024; Received in revised form 29 October 2024; Accepted 12 November 2024
Available online 17 November 2024
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their ability to dissolve lignin and carbon dioxide (CO_2) is discussed. Our method thus guides the design of effective solvents with optimal Kamlet-Taft parameter values dissolving and converting biomass and CO_2 into valuable chemicals.

1. Introduction

Over the past few decades, there has been a growing emphasis on the development of cost-effective and non-toxic bioderived (environmentally friendly) solvents which have the potential to be used in various chemical and pharmaceutical industries for developing sustainable chemical processes and products [1–3]. These efforts are especially focused on replacing solvents which constitute approximately 80% of industrial solvents that cause significant environmental contamination issues due to their toxicity and volatility with green alternatives [4]. In recent years, various environmentally friendly alternatives to traditional volatile organic solvents have emerged; these include aqueous systems [5], sub- and supercritical solvents [6,7], ionic liquids (ILs) [8,9], and deep eutectic solvents (DESs) [10].

ILs and DESs are classes of green solvents with recent research due to their attractive properties and potential use as solvents in many industrial applications such as biomass processing [11,12], carbon capture [13,14], as well as electrolytes in batteries and capacitors, separations, material synthesis, etc. [15,16]. By the optimal selection of specific ionic pairs in ILs and hydrogen bond acceptors (HBAs)/hydrogen bond donors (HBDs) pairs in DESs, ILs and DESs have the intriguing potential to serve as "designer" solvents, with optimized physicochemical properties for specific industrial tasks [15,16]. While ILs are salts with relatively low melting temperatures (below 100 °C), with a vast majority of them are in a liquid state in room temperatures [8], DESs are prepared by a mixture of Lewis or Brønsted acids and bases that act as HBDs or HBAs. Physically mixing two or more such compounds at specific molar ratios results in a DES whose liquid phase is a result of the strong interactions such as hydrogen bonding, electrostatics, and van der Waals (vdW) between HBAs and HBDs [17]. DESs offer a significant advantage over ILs, primarily due to their cost-effective and easy preparation without additional purification or need of costly/toxic solvents [18,19].

For these designer solvents to be utilized in novel industrial applications, one of the most important thermophysical classes of characteristics that must be understood is their solvation properties, in particular their polarity [20]. Polarity plays a key role in governing the interactions between a solvent and a dissolved substance [20], and solvation polarity influences reaction rate mechanisms and product yields.

The two most commonly used scales to characterize solvent polarity in terms of solvatochromic parameters are the Dimroth and Reichardt $E_T(30)$ scales [21]. However, these can only provide qualitative trends. The $E_T(30)$ polarity scale, which measures solvent polarizability, is strongly influenced by the solvent's capacity to donate and accept hydrogen bonds [21]. In contrast, Kamlet and Taft introduced a multi-parameter scale based on solvatochromic probes to evaluate solvent acidity (α), which describes the solvent's hydrogen-bond donating capability; basicity (β), which represents the hydrogen-bond accepting ability; and dipolarity/polarizability (π^*), indicating the solvent's dipolar and polarizability characteristics [22–24].

The Kamlet-Taft parameters are experimentally determined utilizing Reichardt's dyes, including 4-nitrobenzenamine, diethyl-(4-nitrophenyl) amine, and 2,6-dichloro-4-(2,4,6-triphenylpyridin-1-ium-1-yl)phenolate [20]. Nonetheless, the cost of these dyes is expensive. Despite the usefulness of these solvatochromic parameters, Kamlet-Taft parameters remain unavailable for a wide range of conventional (organic solvents) and non-conventional solvents (ILs and DESs). The complexity of ILs/-DESs pose challenges in measuring their solvatochromic parameters. For instance, the limited solubility of Reichardt's dye in aqueous solutions of ILs/DESs hinders precise measurement of their α and β [25].

To address this challenge, various computational and experimental techniques have been devised to predict the solvatochromic parameters of ILs/DESs [25–28]. For ILs, Cláudio et al. (2014) [29] and Kurnia et al. (2015) [30] used the conductor-like screening model for real solvents (COSMO-RS) to calculate IL acidity and basicity. Further, Venkatraman and Lethesh (2019) [27] developed multilinear regression (MLR) models for calculating Kamlet-Taft parameters of 208 ILs. In addition to ILs, the calculation of Kamlet-Taft parameters of DESs has also been investigated. Madeira et al. (2017) [31] and Teles et al. (2017) [32] developed a technique to calculate the Kamlet-Taft parameters of DESs based on NMR spectroscopy. However, the data for method validation is limited. Recently, Kundu et al. (2020) [33] and Wojeicchowski et al. (2021) [25] reported a COSMO-RS-based quantitative structural-property relationship (QSPR) model to calculate the Kamlet-Taft parameters of DESs. The results are promising, but the developed models are only for specific HBA and HBD families. Kundu et al. (2020) [33] developed several QSPR models again for specific HBA families, such as a separate model for each of HBAs, *i.e.*, choline chloride ([Ch]Cl), thymol, DL-menthol, and betaine. Similarly, Wojeicchowski et al. (2021) [25] developed QSPR models for specific HBD families, such as a separate model for acidic and non-acidic HBDs. The limited dataset sizes in the above studies pose challenges for

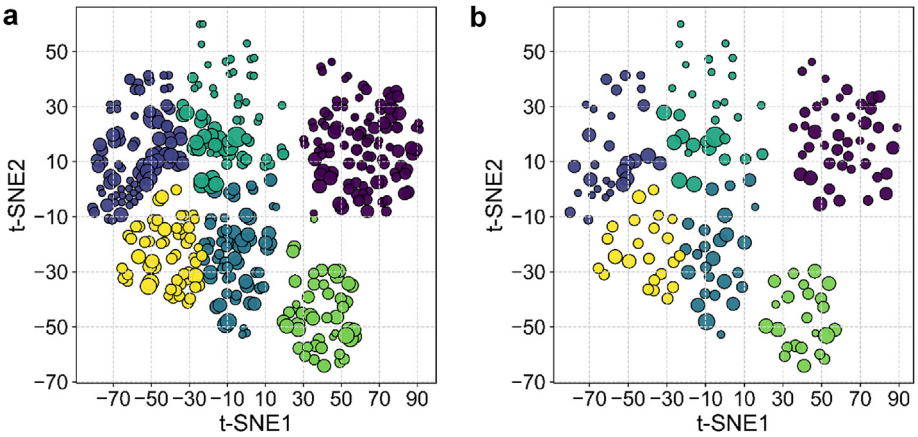


Fig. 1. Visualization of structural representations of two-dimensional t-SNE plot for organic compounds: (a) present study chemical space and (b) literature reported solvents chemical space. For Fig. (b), the literature data was taken from Wojeicchowski et al. [25].

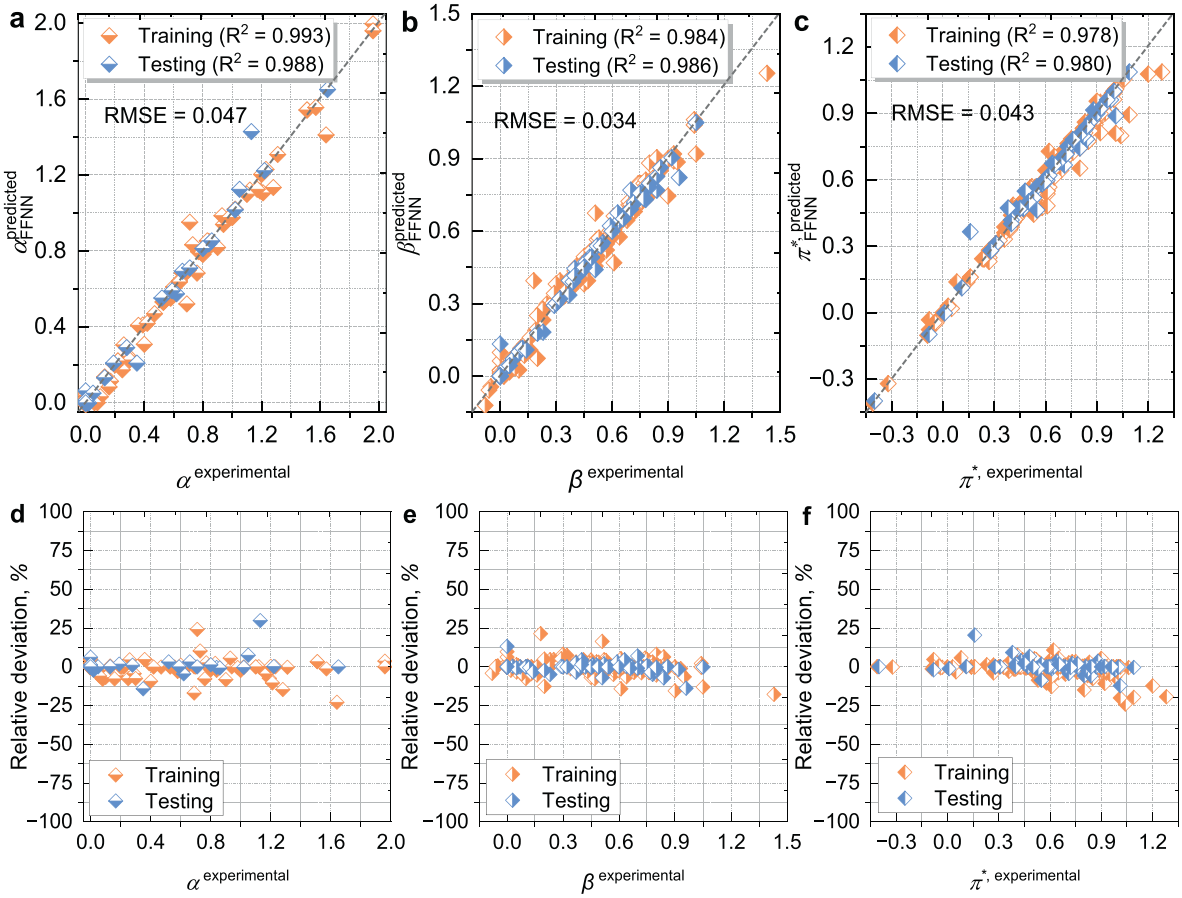


Fig. 2. FFNN predicted Kamlet-Taft parameters of organic compounds: (a) α, (b) β, and (c) π^*. Relative deviation between the experimental and predicted Kamlet-Taft parameters: (d) α, (e) β, and (f) π^*.

the transferability of literature reported models to unseen data.

State-of-the-art green solvents and their growing applications in separation and purification processes, biomass processing, carbon capture, battery research, etc. are of utmost importance; therefore, construction of polarity charts for them is critical. The present study focuses on the prediction of Kamlet-Taft parameters (α, β, and π^*) of designer solvents (organic solvents, ILs, and DESs) using COSMO-RS-derived molecular descriptors. Initially, we compiled a large data set of Kamlet-Taft parameters comprising 293 data points for organic solvents, 452 for ILs, and 328 for DESs. Subsequently, we employed COSMO-RS-derived molecular descriptors to develop two different ML models as well as conducted a comparative analysis of solvatochromic parameters with existing models reported in the literature.

Table 1
Performance metrics of different solvent Kamlet-Taft parameters for an FFNN model.

Solvent type	Property	Data set	Number of data points	R^2	RMSE	*p*-value
Organic solvents	α	Training	225	0.993	0.100	< 0.0001
		Testing	56	0.988	0.047	< 0.0001
	β	Training	234	0.984	0.037	< 0.0001
		Testing	59	0.986	0.034	< 0.0001
	π^*	Training	209	0.978	0.043	< 0.0001
		Testing	52	0.980	0.043	< 0.0001
ILs	α	Training	311	0.884	0.087	< 0.0001
		Testing	77	0.951	0.060	< 0.0001
	β	Training	350	0.961	0.066	< 0.0001
		Testing	87	0.983	0.042	< 0.0001
	π^*	Training	334	0.757	0.064	< 0.0001
		Testing	83	0.928	0.034	< 0.0001
DESs	α	Training	246	0.966	0.062	< 0.0001
		Testing	61	0.983	0.043	< 0.0001
	β	Training	248	0.961	0.065	< 0.0001
		Testing	62	0.977	0.048	< 0.0001
	π^*	Training	254	0.962	0.054	< 0.0001
		Testing	64	0.995	0.021	< 0.0001

2. Methodology

2.1. Solvent Kamlet-Taft parameters database

Kamlet-Taft empirical polarity scales for hydrogen-bond acidity (α), basicity (β), and dipolarity-polarizability (π^*) were compiled from literature for the dye sets Reichardt's dye, N,N-diethyl-4-nitroaniline, and 4-nitroaniline. A comprehensive set of 293, 452, and 328 data points were collected from existing literature on organic compounds, ILs, and DESs, respectively. For organic compounds, 175 data points were taken from Marcus (1993) [34], Mohan et al. (2022) [35], and Wojeicchowski et al. (2021) [25], and the remaining 118 data points were collected from the literature.

For ILs, 208 data points were taken from Venkatraman and Lethesh (2019) [27], and the remaining 244 data points were compiled from published literature to ensure that the data set is up to date and covers a wide chemical space. The total updated number of data points were 452 for 414 different ILs at different temperatures. The dataset comprised 135 unique cations and 82 unique anions.

For DESs, a total of 328 data points were assembled from the literature on 313 different DESs at different molar ratios (19:1 to 1:15) and

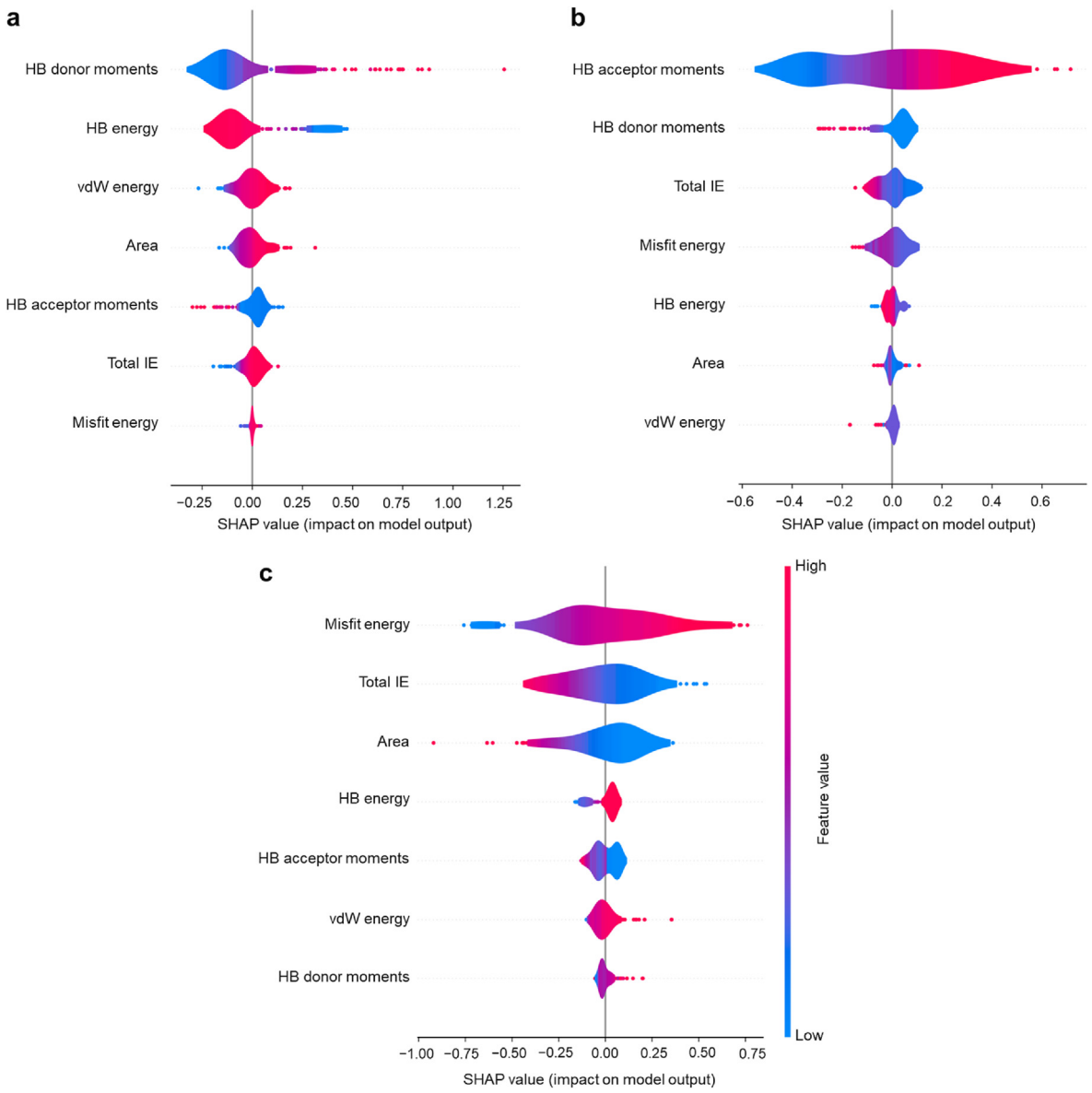


Fig. 3. SHAP feature importance of organic compounds for Kamlet-Taft parameters: (a) α, (b) β, and (c) π^*.

temperatures (298.15 K–353.15 K). In total, 88 data points were taken from Kundu et al. (2020) [33] and Wojeicchowski et al. (2021) [25], and the remaining 240 data points were compiled from literatures. We ensured that the DES Kamlet-Taft parameters data are unique (*i.e.,* there are no duplicate data points). The data set encompasses wide HBA and HBD structural diversity, with 44 HBAs and 76 HBDs. Some common HBAs examined here are metal chlorides ($ZnCl_2$, $FeCl_3$, $CuCl_2$, $AlCl_3$, etc.) as well as ammonium-based ([Ch]Cl, [Ch][Lev], [TBA]Cl, [TBA]Br, [TEA]Cl, [TMA]Ace], betaine, etc.), superbase-based (guanidine chloride, etc.), imidazolium-based ([EMIM]Cl, [BMIM]Cl, [AMIM]Cl, [MIM]Cl, etc.), and non-ionic compounds (menthol, thymol, L-proline, L-alanine, matrine, etc.), and HBDs are sugars, amines, acids, phenols, and alcohols. It is worth noting that the database compiled in this study is significantly larger compared to the Kamlet-Taft parameter databases utilized in previous studies [25,27,33–35]. This study not only compiles a large dataset for designer solvents but also explores a wide range of structural diversity to develop accurate models for predicting Kamlet-Taft parameters.

2.2. *Input feature generation using the COSMO-RS*

In principle, the performance of ML models relies on several key factors, including the dataset, the utilization of ML algorithms, and the selection of input features [36,37]. Among these factors, the selection of suitable input features plays a significant role in determining the accuracy of ML models. Furthermore, input features should have physical significance to aid in discovering the underlying principles of the model and achieving interpretability of the predictive model. In our recent investigations, we demonstrated the reliability of COSMO-RS-derived molecular features as inputs for developing highly accurate machine learning models. These models were trained to predict properties of ILs and DESs as well as CO_2 solubility in DESs [1–3,13,14,38]. We also have demonstrated the use of COSMO-RS molecular descriptors to calculate the basicity (β) of organic compounds [35].

The details of molecular geometry optimization and further COSMO-RS calculations are discussed in the Supporting Information (section S1). For each molecule, the following molecular descriptors were calculated

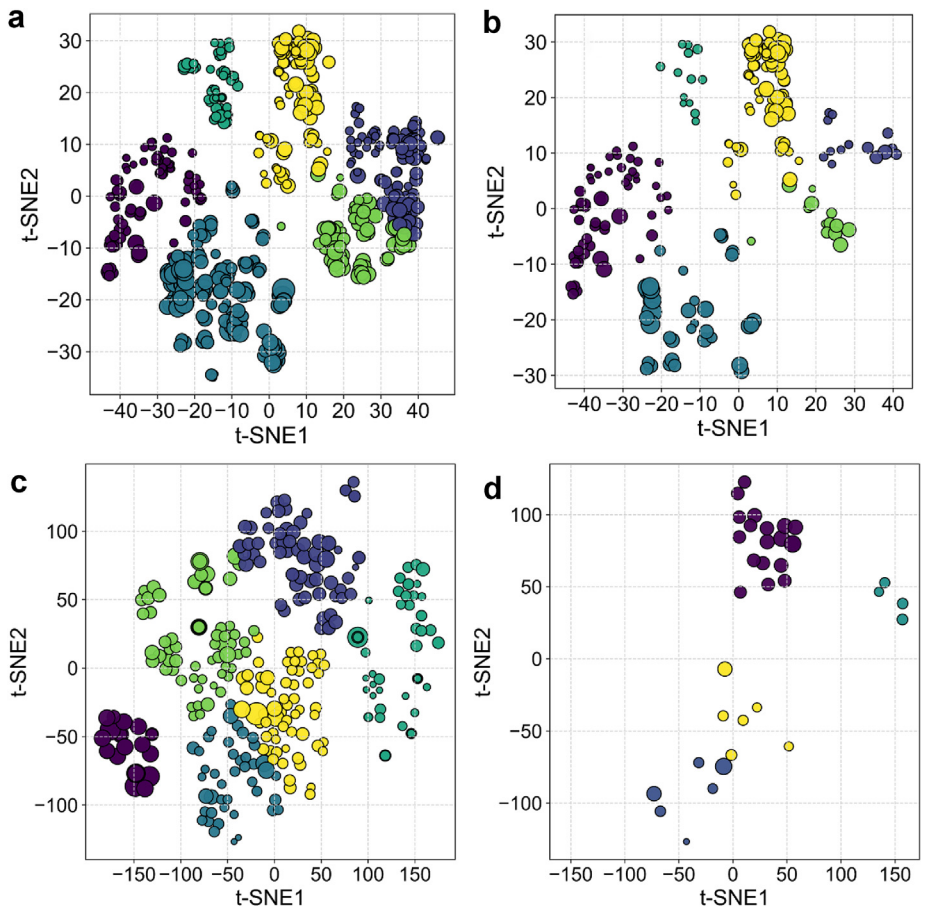


Fig. 4. Visualization of structural representations of two-dimensional t-SNE plot for ILs: (a) present study chemical space and (b) literature reported solvents chemical space. For Fig. (b), the literature data was taken from Venkatraman and Lethesh [27]. DESs: (c) present study chemical space and (d) literature reported solvents chemical space. For Fig. (d), the literature data was taken from Wojeicchowski et al. [25].

using COSMO-RS: (1) total interaction energy (total IE), (2) misfit energy, (3) hydrogen bonding energy (HB energy), (4) van der Waals energy (vdW energy), (5) HB acceptor moments, (6) HB donor moments, and (7) molecular surface area. The moments of donors and acceptors in the COSMO-RS represent the distribution of electron density around potential HB sites on the molecular surface. A higher donor moment indicates a stronger ability to act as HB donor, while a higher acceptor moment suggests a stronger ability to act as HB acceptor. The explicit interactions between cations and anions in ILs, as well as between the HBA and HBD in DESs, are not calculated due to the high computational cost. To calculate input features of ILs/DESs, instead of modeling the complex structures of ILs and DESs, we have simplified the approach by defining these solvents based on their molecular compositions and stoichiometries. For ILs, we modeled the cation and anion separately and combined their features in an equimolar combination to represent IL.

In the case of DESs, the process is different because DESs are composed of an HBA and HBD in specific molar ratios, which are crucial to their properties. We calculate the COSMO-RS descriptors for the HBA and HBD separately and then combine them based on their molar ratio. For instance, in a DES such as ChCl/urea with a 1:2 molar ratio, the COSMO-RS descriptors for ChCl and urea are calculated individually and then merged, with 1 mol of ChCl and 2 mol of urea, to represent the overall descriptor for the DES. This approach reflects the stoichiometry of the DES and captures the contributions of both components to properties. It is also important to mention that this methodology aligns with our previous work, in which we applied a composition-weighted average for predicting CO_2 solubility in DESs and achieved excellent predictive performance [1,13,14]. Similarly, for ILs, we calculated input features for anion and cation independently and developed ML models for IL properties and achieved excellent prediction with high accuracy [2,3,38].

3. Machine learning models

We developed a feed-forward neural network (FFNN)–based ML model to predict the Kamlet-Taft parameters of designer solvents using the JMP Pro statistical software (JMP® Pro SAS 17.2.0) [39]. We utilized the COSMO-RS-derived molecular descriptors and temperature as input features. In training the FFNN model, a nonparametric function f is sought to map the input features to the predicted Kamlet-Taft parameter, as shown below [13]:

$$Kamlet-Taft = f\left(\begin{matrix} T, total\ IE, misfit, HB\ energy, vdW\ energy, \\ HB\ acceptor\ and\ donor\ moments, A \end{matrix}\right) \quad (1)$$

where T is the temperature (K), A is the molecular surface area of solvents. The FFNN model was developed with 2 hidden layers and 20 neurons in each hidden layer and a fixed learning rate of 0.1. The number of tours was set to 10,000, and optimized using a squared penalty method, while the remaining hyperparameters in the JMP® Pro SAS 17.2.0 software were set to default [39]. The principle of the FFNN algorithm were discussed in detail in our previous study [11,13,14].

To evaluate the predictive accuracy of ML models, various performance metrics were computed, such as the determination coefficient (R^2), the root mean square error (RMSE), the absolute annual relative deviation (AARD), and the p-value of model significance. The optimal ML model was determined as that with the lowest RMSE and highest R^2

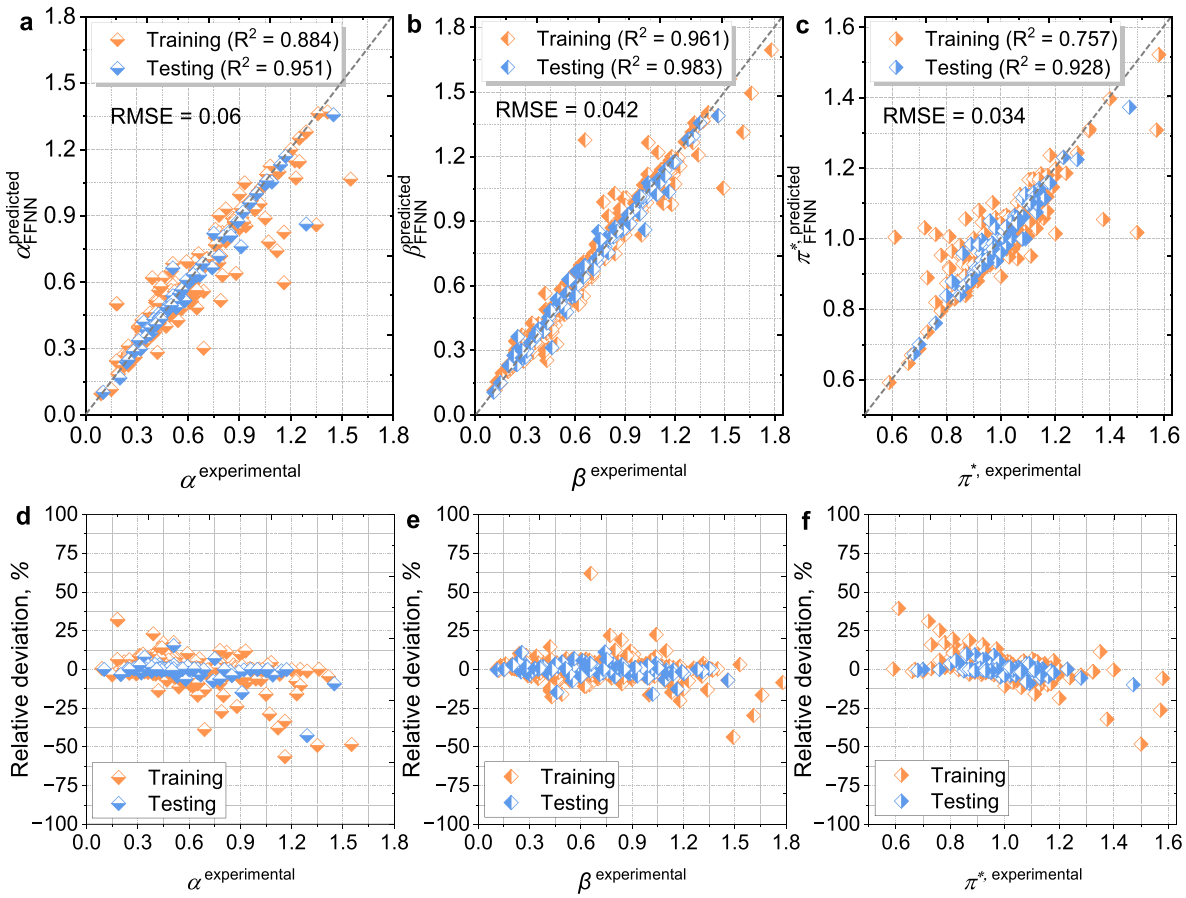


Fig. 5. FFNN predicted Kamlet-Taft parameters of ILs: (a) α, (b) β, and (c) π^*. Relative deviation between the experimental and predicted Kamlet-Taft parameters: (d) α, (e) β, and (f) π^*.

values [3,14]. Furthermore, SHapley Additive exPlanations (SHAP) analysis was performed to interpret the importance of input features [40]. SHAP, a Python-based tool, was utilized to interpret the results of the ML models. More details about SHAP analysis are provided in our recent publications [3,14,41].

4. Results and discussion

4.1. Chemical space analysis of organic solvents

Describing the diversity of chemical space is a complex task, which is difficult to conceptualize and even more challenging to analyze computationally. The t-distributed stochastic neighbor embedding (t-SNE) is a dimensionality reduction approach to maintaining proximity relationships between data points as they transition from high dimensionality to a lower-dimensional sub-space [42,43]. We utilized the t-SNE technique implemented in the scikit-learn library to reduce the dimensionality of the data [44]. To calculate the chemical space of investigated solvents, we used the Morgan fingerprints with a radius of 3 and a vector length of 2048 [45,46]. Fig. 1 shows the visual representation of the chemical space derived from our study against the solvent's chemical space reported in the literature. The k-means clustering method was used to visualize the distribution of organic solvents clusters [47]. The optimal number of clusters in the dataset was calculated using the Elbow method, and six clusters were found in the dataset [48]. As shown in Fig. 1, our study covers larger structural diversity of organic solvents as compared with literature reported solvents. The richness of our dataset is evident with over 290 samples, reflecting a comprehensive spectrum of organic compounds across six different clusters, contrasted with the solvent space of 175 data samples drawn from existing literature sources [25]. Therefore, leveraging the comprehensive and diverse dataset, we train ML algorithms to better capture diversity of organic solvent categories.

4.2. ML models for organic solvent Kamlet–Taft parameters

We first developed two different ML models to predict the Kamlet–Taft parameters for conventional organic compounds using the COSMO-RS-derived molecular input features. As discussed earlier, we assembled Kamlet–Taft parameters for 293 organic compounds, which is the most extensive experimental dataset of Kamlet–Taft parameters to date for organic compounds. The model was trained using the 80% of data and the remaining 20% of data utilized for testing. To develop the ML models, we used the COSMO-RS-derived input features: the total IE, misfit energy, HB energy, vdW energy, HB acceptor moments, HB donor moments, and molecular surface area. First, we trained a MLR model with COSMO-RS-derived input features, and the results are illustrated in Fig. S1 (Supporting Information). The MLR model underestimates the solvatochromic empirical polarity scales for β and π^* with R^2 values for the training and testing sets of 0.635–0.741 and 0.409–0.676, respectively. In contrast, the MLR model predictions for the acidity (α; Fig. S1a) parameter were satisfactory, with R^2 values of 0.901 and 0.856 for training and testing, respectively.

To improve the Kamlet–Taft empirical polarity parameter predictions, we developed an FFNN model using the same input features. As depicted in Fig. 2a-c, the predictions of Kamlet-Taft parameters with FFNN are in excellent agreement with experiment with high R^2 (0.980−0.988) and low RMSE (0.034−0.047) values on the testing set (Table 1). The distributions of the relative deviations of the parameters with respect to experiment are shown in Fig. 2(d–f). The majority of the predicted data lie within 10% of AARD, with a few data points lying

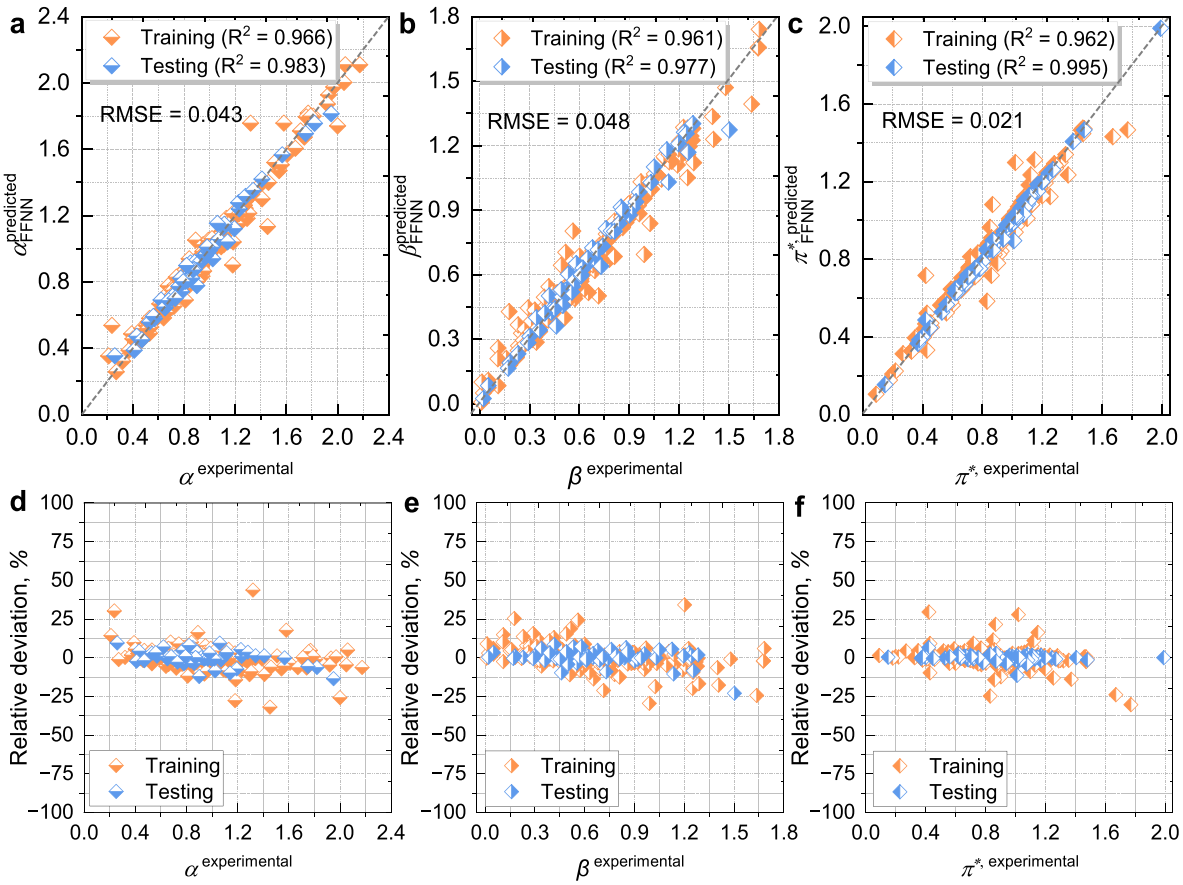


Fig. 6. FFNN predicted Kamlet-Taft parameters of DESs: (a) α, (b) β, and (c) π^*. Relative deviation between the experimental and predicted Kamlet-Taft parameters: (d) α, (e) β, and (f) π^*.

beyond 20%. These results clearly demonstrate the high accuracy of the developed ML model for solvatochromic parameter prediction. In recent work, Wojeicchowski et al. (2021) [25] reported a QSPR model for predicting the solvatochromic parameters of organic compounds using COSMO-RS-derived features and using 175 data points, yielding R^2 values of 0.86, 0.63, and 0.42 for α, β, and π^*, respectively. The performance metrics achieved by Wojeicchowski et al. (2021) [25] are comparable to those of our MLR model.

Next, the FFNN model results were interpreted using SHAP analysis to extract the feature importance (Fig. 3). For each feature, SHAP > 0 implies that the model uses that feature to raise a Kamlet–Taft parameter, while SHAP < 0 lowers the values [3,14]. For hydrogen-bond acidity (α), the features HB donor moments, HB energy, vdW energy, and molecular surface area are the most important (Fig. 3a). The SHAP analysis suggests that for solvents with high HB donor moments, the acidity tends to increase, which is in agreement with the experimental measurements [23]. A molecule that has a high HB donor moment can act as a strong HB donor, *i.e.*, high acidity, α. Furthermore, a molecule with strong HB energy (more negative, *i.e.*, blue region) corresponds to a high HB donor moment, which again is in agreement with experimental measurements [25,31]. Lower vdW energy (more positive, *i.e.*, pink region) correlates with higher α. For the β, the features, HB acceptor moments, HB donor moments, total IE, and misfit energy are the most important (Fig. 3b). As expected, higher HB acceptor moments tend to increase basicity, and the SHAP value is correspondingly positive (SHAP > 0), which again is in agreement with the experiment [24]. Furthermore, lower SHAP values of HB donor moments, total IE, and misfit energies correspond to higher basicity, again in agreement with experimental observations [49]. In the case of polarizability (π^*), the misfit energy, total IE, area, and HB energy are the most influential features (Fig. 3c). Higher misfit energy corresponds to higher polarizability. Generally, misfit electrostatic energy arises from the polar charges of the molecule. Also, lower molecular surface area and stronger HB energies contribute to increased molecular polarizability. For example, the surface area of a water molecule is low, and HB energy is higher, which results in higher polarizability ($\pi^* = 1.2$).

4.3. ML model for designer solvent Kamlet–Taft parameters

There have been several attempts to predict Kamlet-Taft parameters for designer solvents such as ILs and DESs. Palomar et al. (2010) [50], Cláudio et al. (2014) [29], and Kurnia et al. (2015) [30] developed QSPR models to predict α and β for ILs, but no studies were conducted for π^*. Also, the number of data points and structural diversity was lower than here. Further, Kundu et al. (2020) [33] and Wojeicchowski et al. (2021) [25] developed several QSPR models to predict the Kamlet-Taft parameters of DESs using the COSMO-RS-derived molecular descriptors for various classes of DES based on separate models for HBAs ([Ch]Cl, [TBA]Br, [TBA]Cl, menthol, and betaine) and HBDs (acidic and non-acidic). Developing multiple QSPR models for each HBA and HBD compound was not transferable on the new data set, making it challenging to derive a generic model.

Using a different approach, we aimed to develop a generalized method for predicting Kamlet-Taft parameters of ILs and DESs. Separate models were constructed for ILs and DESs. Before developing ML models, we performed chemical space and clustering analysis for ILs and DESs with a t-SNE method on the molecular fingerprints of all ILs and DESs, and the results are compared with literature reported ILs/DESs [25,27]. The chemical space analysis of ILs is shown in Fig. 4a and b. As discussed earlier, the optimal cluster size was determined using the Elbow method, and the k-means clustering method was utilized for clustering

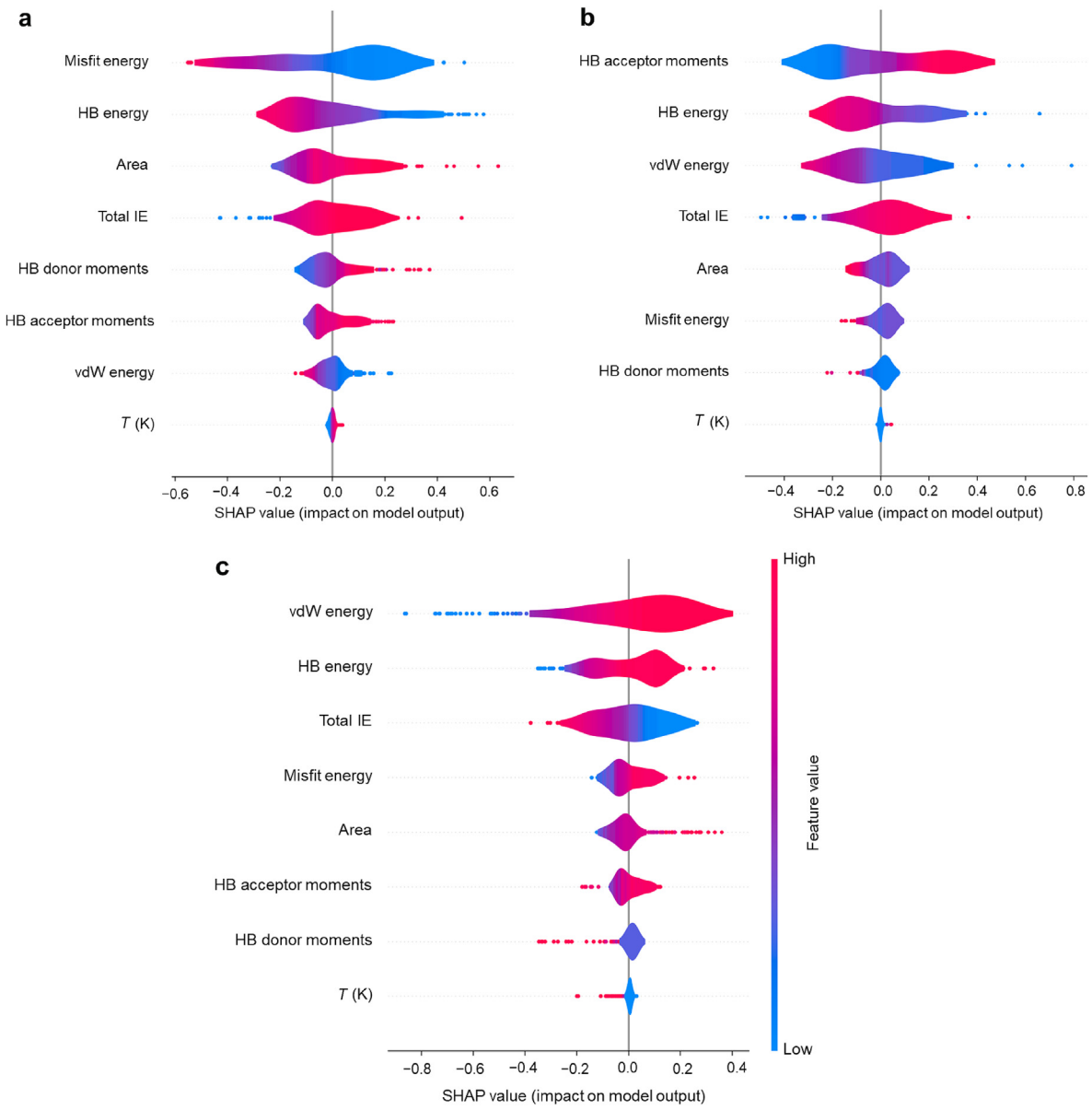


Fig. 7. SHAP feature importance of ILs Kamlet-Taft parameters: (a) α, (b) β, and (c) π^*.

visualization [47,48]. For ILs (Fig. 4a and b), except for deep purple– and yellow-colored clusters, the population of data points in remaining clusters was much less in the literature reported models. For example, the data points in the Caribbean current-, green-, and violet-colored clusters were very rich in the present study as compared to the literature ILs. Additionally, the chemical space analysis of DESs was performed and is shown in Fig. 4c and d. In the present study, the coverage of DESs structures was larger with a wide range of t-SNE values in the chemical space with more clusters (Fig. 4c), whereas the chemical space coverage of literature reported DESs was very small with a limited number of DESs (Fig. 4d). It is worth mentioning that the dataset of literature reported Kamlet-Taft parameters is smaller than our IL/DES dataset, covering structures ranging from 2 to 100 times larger. Due to the fact that previous predictive model studies had limited coverage of chemical space compared to ours, the transferability of their models to new datasets may be limited and sub-optimal [25,27,33]. Not only did the present study compile a comprehensive dataset of Kamlet-Taft parameters for ILs and DESs; it also covered a broader range of chemical space (Fig. 4).

We developed the MLR model to predict the solvatochromic parameters of ILs and DESs using the COSMO-RS-derived molecular input features. The input features comprise the temperature (T (K)), total IE, misfit energy, HB energy, vdW energy, HB acceptor moments, HBA to HBD molar ratio (MR only for DES), and HB donor moments. As mentioned above, 80% of the dataset was utilized for training, and 20% of the data was used for testing. The number of data points splitting for training and testing is given in Table 1. As shown in Figs. S2 and S3, the MLR model underestimates the α, β, and π^* of ILs and DESs with R^2 values of 0.233–0.760 and 0.215–0.771 for the training and testing sets, respectively. It is worthwhile to mention that most of ILs have been shown to be highly viscous, and to avoid high viscosity related problems, researchers have used a cosolvent method to reduce the viscosity of ILs and achieve better results [51,52]. There is a large number of data points available for properties of IL-cosolvent mixtures, such as viscosity, density, activity, and surface tension [53], but not enough datasets for Kamlet-Taft parameters for mixtures (ILs and co-solvents). Thus, the present study is not focused on the IL-cosolvent mixtures.

Fig. 5 illustrates the correlation of experimental and FFNN-predicted Kamlet-Taft parameters for ILs. The FFNN model predictions are again in excellent agreement with experiment and are superior to the MLR model. As illustrated in Fig. 5(a–c), the test set results show more accurate

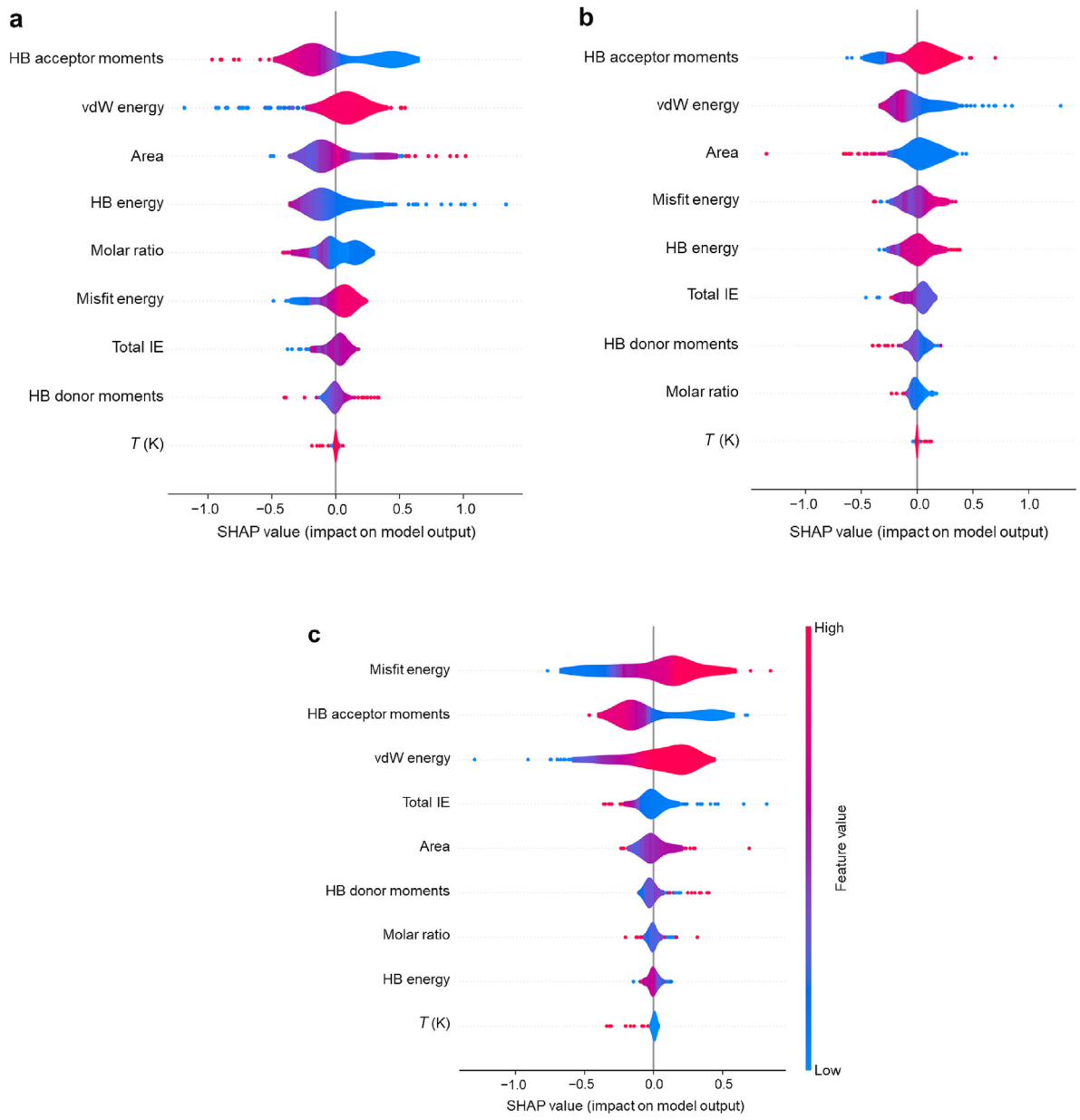


Fig. 8. SHAP feature importance of DESs Kamlet-Taft parameters: (a) α, (b) β, and (c) π^*.

predictions compared to the training set results, and the training set exhibits significant disturbances in the high-value regions of α, β, and π^*. These large deviations arise from the complexity of the structures and molecular interactions of ILs (*i.e.*, cations and anions). Additionally, the observed disturbances in the training set could also stem from variability in the experimental measurements and potential impurities of the ILs that can affect results. Since these properties are highly sensitive to experimental measurements and methods used, slight variations in conditions (such as temperature, solvent purity, or instrument calibration) can lead to discrepancies in the reported values. Notwithstanding, as compared to existing literature reported models for the prediction of Kamlet-Taft parameters of ILs, the present model provides a more accurate prediction of α, β, and π^* of ILs together with a larger number of data points and larger covered chemical space for ILs. The relative deviations of the Kamlet-Taft parameters were plotted against the experimental values in Fig. 5(d–f). The majority of the test data set lies within 15% of AARD, while for the training set, the RD dataset lies within 25% of AARD.

Fig. 6(a–c) shows the prediction of Kamlet-Taft parameters of DESs using the FFNN model. The agreement with experiment is excellent for both the training and testing datasets. The performance metrics (R^2 = 0.961–0.995 and RMSE = 0.021–0.065) of α, β, and π^* indicate high accuracy (Table 1). It is important to note that the present study develops a generic model for DES Kamlet-Taft parameter predictions, and it is more advanced than that of Kundu et al. (2020) [33] and Wojeichowski et al. (2021) [25] in which individual HBA and HBD QSPR models were developed. The relative deviations of Kamlet-Taft parameters relative to the experimental values are shown in Fig. 6(d–f). The majority lie within 10% of AARD with a few data points beyond 20%. These results again clearly demonstrate the accuracy of the developed ML model for DESs Kamlet-Taft parameter predictions. It is important to mention that the variations of R^2 and RMSE values across solvent types (organic solvents, ILs, and DESs) in Table 1 are not random. They are primarily driven by differences in structural complexity, interaction types, and non-linearities in solvent-property relationships. The molecular structures of organic solvents typically are not as complex as ILs and DESs and are more easily captured by ML models, while ILs are more structurally and electrostatically complex. This complexity is reflected in lower R^2 and higher RMSE values, especially for properties such as α and

Table 2
Comparison of Kamlet-Taft prediction models for organic solvents, ILs, and DESs.

Solvent type	Property	Method	No. of Data points	R^2	RMSE	Reference
Organic solvents	E^N_T	MLR	70	0.960	–	[50]
Organic solvents	β	MLR	175	0.919	–	[49]
	π^*	MLR	175	0.800	–	[49]
Organic solvents	α	MLR	175	0.860	–	[25]
	β	MLR	175	0.630	–	[25]
	π^*	MLR	175	0.420	–	[25]
Organic solvents	α	FFNN	281	0.988	0.047	Present study
	β	FFNN	293	0.986	0.034	Present study
	π^*	FFNN	261	0.980	0.043	Present study
ILs	E^N_T	MLR	60	0.950	–	[50]
ILs	β	MLR	21	0.923	–	[29]
ILs ($[NTf_2]^-$)	α	MLR	17	0.944	–	[30]
ILs	α	MLR	209	0.42	0.20	[27]
		CTREE[a]	209	0.69	0.18	[27]
	β	MLR	229	0.87	0.13	[27]
		CTREE[a]	229	0.87	0.15	[27]
	π^*	MLR	208	0.28	0.11	[27]
		CTREE[a]	208	0.54	0.11	[27]
ILs	α	FFNN	388	0.951	0.060	Present study
	β	FFNN	437	0.983	0.042	Present study
	π^*	FFNN	417	0.928	0.034	Present study
DESs ([Ch] Cl)	α	MLR	19	0.990	–	[33]
	β	MLR	19	0.920	–	[33]
DESs (DL-Menthol)	α	MLR	11	0.830	–	[33]
	β	MLR	11	0.893	–	[33]
DESs (betaine)	α	MLR	3	0.966	–	[33]
	β	MLR	3	0.898	–	[33]
Acidic DESs	α	MLR	27	0.797	–	[25]
	β	MLR	27	0.816	–	[25]
	π^*	MLR	27	0.831	–	[25]
Non-acidic DESs	α	MLR	17	0.867	–	[25]
	β	MLR	17	0.707	–	[25]
	π^*	MLR	17	0.823	–	[25]
DESs	α	FFNN	307	0.982	0.043	Present study
	β	FFNN	310	0.977	0.048	Present study
	π^*	FFNN	318	0.995	0.021	Present study

[a] CTREE: conditional inference tree-based analysis.

π^*. DESs themselves fall in between organic solvents and ILs in terms of structural complexity. As we discuss, DESs generally consist of HBD and HBA, creating more complex interactions than organic solvents but less structural diversity than ILs. Furthermore, the data availability and distribution also significantly affect the R^2 and RMSE values.

Furthermore, SHAP was performed to extract feature importance and facilitate the rational design of ILs and DESs. Fig. 7 shows the feature importance for IL Kamlet-Taft parameters (α, β, and π^*). As shown in Fig. 7a, the misfit energy, HB energy, area, total IE, and HB donor moments are the most important features for acidity (α). High HB donor moments, surface area, and stronger total IE tends to increase α. An increase in the HB acceptor moments and stronger HB energies result in higher basicity β of ILs (Fig. 7b). For example, as the HB acceptors moment of the cation increases from 1-ethyl-3-methylimidazolium to 1-butyl-3-methylimidazolium ($[EMIM]^+$ to $[BMIM]^+$) with $[NTf_2]^-$, the basicity of the IL increases. Similarly, as the HB acceptor moment of anions increases, the basicity of the ILs also increases, in line with the experimental measurements [29]. For π^*, weaker vdW energies, stronger total IE, and higher HB acceptor moments lead to higher polarizability (Fig. 7c).

Fig. 8 shows the SHAP feature importances for DESs. Low HB acceptor moments, weak vdW energy, and large HB donor moments predict high acidity (α) of DESs. For basicity, HB acceptor moments, vdW energy (*i.e.*, more negative), and HB energy are the most important features. DESs that possess higher HB acceptor moments tend to have enhanced basicity, in agreement with experiment [32]. Furthermore, weaker misfit energy, lower HB acceptor moments, and stronger vdW energies are the most important features for higher polarizability (π^*) (Fig. 8c). Notably, temperature is the least important in all the predictions, as illustrated in Figs. 7 and 8. This could be attributed to the fact that the IL/DES dataset has limited variation in temperature data, with over 91% of the data recorded at 298.15 K.

4.4. Comparison with literature reported models

It is important to compare our models with literature reported models (Table 2). We use RMSE and R^2 as the performance metrics for the comparisons. A significant difference from other studies is that our study compiled a more extensive dataset with a wider range of unique solvents (293 organic compounds, 414 ILs, and 313 DESs) at different temperatures and molar ratios. Palomar et al. (2010) [50] developed an MLR model based on the COSMO-RS-derived sigma profiles to predict the empirical solvent polarity (E^N_T) scale of 70 organic compounds, and the model resulted in a good R^2 value of 0.960. Further, Sherwood et al. (2019) [49] and Wojeicchowski et al. (2021) [25] developed QSPR models to predict the Kamlet-Taft parameters of 175 organic compounds using the COSMO-RS-derived molecular descriptors. The performance of these MLR models on the Kamlet-Taft parameters was weaker as compared to the FFNN model in the present study (R^2 = ~0.99).

For ILs, Palomar et al. (2010) [50], Cláudio et al. (2014) [29], and Kurnia et al. (2015) [30] developed QSPR models to predict E^N_T and Kamlet-Taft parameters with limited data sets (17–60 data points). Further, Venkatraman and Lethesh (2019) [27] developed an MLR and conditional inference tree–based analysis (CTREE) model with 209 data points to predict the Kamlet-Taft parameters, reporting R^2 values of acidity and polarizability of 0.42–0.69, which is lower than present study results. Venkatraman and Lethesh (2019) [27] obtained better results for basicity (R^2 = 0.87); however, this is lower than that of the present study (R^2 = 0.983). For DESs, Kundu et al. (2020) [33] and Wojeicchowski et al. (2021) [25] developed many QSPR models, for each individual DES component (*i.e.*, HBA and HBD categories) and obtained less accurate models for Kamlet-Taft parameters than reported here. These studies used far smaller numbers of data points (3–27 DESs) for model development–11 to 100 times lower than the present DES data set. As compared to reported methods, our predictions with the FFNN model obtained better R^2 values and lower RMSEs (Table 1). A further improvement of the present study is that our models were trained and tested with larger structural diversity and diverse functionality of organic solvents, ILs, and DESs and with a greater number of data points.

4.5. Application of Kamlet-Taft parameters in lignin and CO_2 solubilities

In the previous sections, we demonstrated how COSMO-RS-derived molecular input features can be effectively utilized to develop accurate ML models for predicting the Kamlet-Taft parameters of organic compounds, ILs, and DESs. Recent studies have shown that ILs and DESs are the potential solvents for dissolving lignocellulosic biomass and capturing carbon dioxide (CO_2) with high efficacy [9,11,13,14]. Though extensive research has been conducted on the dissolution of cellulose in ILs/DESs, with a well-documented correlation between cellulose solubility and Kamlet–Taft parameters [54], much less attention has been given to understanding how the Kamlet–Taft parameters influence the solubility of lignin in ILs/DESs. Fig. 9(a–c) examines this relationship for

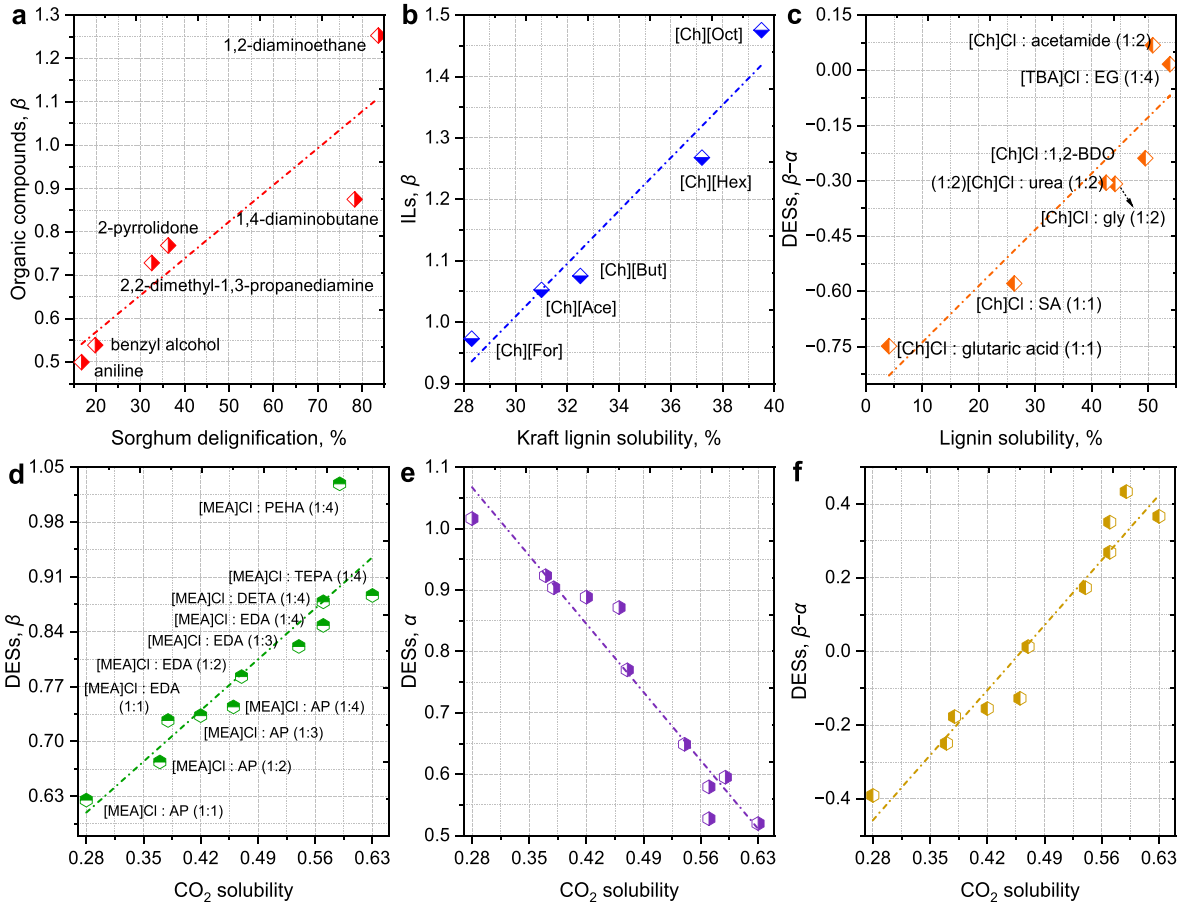


Fig. 9. Relationship between the FFNN predicted Kamlet–Taft parameters of the investigated solvents and the corresponding solubilities of lignin in (a) organic compounds (experimental data taken from Achinivu et al. [55]), (b) ILs (experimental data taken from Hou et al. [56]), and (c) DESs (experimental data taken from Zhang et al. [58]). Panels (d–f) correspond to the solubilities of CO_2 in DESs (experimental data taken from Shukla and Mikkola [59]) with respect to (d) basicity β, (e) acidity α, and (f) β-α.

organic compounds, ILs, and DESs.

Achinivu et al. (2021) [55] recently investigated the delignification of sorghum in various organosolv, including amines and aromatic solvents, and reported that amines are better solvents for high biomass delignification. To understand delignification, we draw a correlation between biomass delignification and Kamlet-Taft parameters of organosolv (Fig. 9a). The results show that the β value of organosolv has a significant correlation with biomass delignification, indicating that solvents with high β tend to have a increased delignification. Additionally, it has been reported that amines exhibit stronger hydrgen bonding accepting capability, enabling them to interact more effectively with lignin, and these observations are in-line with the high β values of amines. Hou et al. (2015) [56] studied the solubility of lignin in cholinium-based ILs with five different carboxylate anions: formate ([For]), acetate ([Ace]), butyrate ([But]), hexanoate ([Hex]), and octanoate ([Oct]). They found that lignin solubility increased with the alkyl chain length of the anions, with the highest solubility observed in [Ch][Oct], and this is correlated with IL basicity, β [56]. A positive correlation was identified between lignin solubility and IL basicity (Fig. 9b). Mohan et al. (2021) [57] performed multiscale molecular simulations to evaluate the dissolution mechanism of lignin in cholinium-based ILs, and reported that as the alkyl chain length of anion increases, stronger interactions, lower activity coefficients, and longer hydrogen bond lifetimes between lignin and anions were observed. These results align with the higher basicity values of ILs. In a recent study, Zhang et al. (2023) [58] reported the solubility of Kraft lignin in various DESs, revealing a positive linear relationship between Kraft lignin solubility and β–α values (Fig. 9c). However, the correlation between Kraft lignin solubility and DES β was weaker than that observed with the β values of organic solvents and ILs. In contrast to ILs, DESs generally exhibit higher α values and slightly lower β and β–α values [58].

Fig. 9(d–f) shows the relationship between CO_2 solubility and DES solvatochromic parameters. We take experimental data from Shukla and Mikkola (2018) [59], who investigated the solubility of CO_2 in different amine-based DESs. The basicity of DESs shows a positive linear correlation with CO_2 solubility, while the acidity of DESs shows a negative correlation. As the molar ratio of [MEA]Cl:AP or [MEA]Cl:EDA increases, the basicity of DESs increases and the acidity decreases, indicating that molar ratio plays significant role in tuning the Kamlet-Taft parameters. Furthermore, a positive and strong linear correlation was obtained between CO_2 solubility and β–α. A positive value of β–α quantifies the basic nature of DESs. Thus, DESs with higher basicity or larger pK_a values possess a greater capability to dissolve CO_2 [59–61]. Qin et al. (2024) [62] quantify the Hammett basicity (H) of DESs using ^{1}H nuclear magnetic resonance (NMR). Hammett basicity mainly focuses on electronic substituent effects on basicity or acidity of aromatic compounds (such as proton transfer or charge-related processes). In contrast, Kamlet-Taft parameters (α and β) deals with the hydrogen bond donating and accepting abilities of a solvent in solvent-solute interactions. Despite these differences, Hammett basicity and Kamlet-Taft parameters correlate well with CO_2 absorption rates, as higher basicity in DESs has been shown to increase CO_2 absorption rates. In systems where CO_2 absorption involves a chemical reaction with a solvent (such as the formation of carbamate or bicarbonate), the Hammett basicity material plays a crucial role. For example, amines, ILs, and DESs with higher Hammett basicity can better react with CO_2, thereby increasing absorption rates. On the

other hand, Kamlet-Taft parameters correlate with CO_2 through non-covalent interactions, such as hydrogen bonding and solubility effects.

5. Conclusions

Accurately predicting Kamlet-Taft parameters for designer solvents is challenging due to the extensive diversity and infinite combinations of the components. The present study investigated ML models for this purpose. We established a diverse dataset for organic solvents (293 data points), ILs (452 data points), and DESs (328 data points) gathered from existing literature. The dataset compiled by the present study is 11–100 times larger than the literature dataset. For ML, COSMO-RS-derived molecular features were calculated and used as input features. Two different ML methods—MLR and FFNN—were employed. Of these, the FFNN model shows better predictive capability than the MLR model, as demonstrated by the performance on the training and testing data (low RMSE and high R^2). The current FFNN model's performance is a great improvement over those reported in the literature. Additionally, we captured the ML model feature importance using SHAP analysis. For solvents, the HB acceptor moment was the most important feature for HB basicity (β) prediction, while higher values of the HB donor moment result in higher HB acidity (α).

Furthermore, for our case studies, we demonstrated the applicability of calculated Kamlet-Taft parameters for lignin and CO_2 solubilities. Strong correlations are found in which solvents with high basicity (β) values have increased lignin and CO_2 solubility, which is in agreement with the experimental observations.

In summary, the proposed ML model has higher accuracy in predicting the Kamlet-Taft parameters over a greater diversity of solvent structures than state-of-the-art models. Furthermore, the FFNN ML model that has been developed in the present study can be directly used for optimizing solvent design to enhance efficiency in lignin processing, CO_2 capture, and doubtless various other applications.

CRediT authorship contribution statement

Mood Mohan: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nikhitha Gugulothu:** Writing – review & editing, Visualization, Methodology, Data curation. **Sreelekha Guggilam:** Writing – review & editing, Visualization. **T. Rajitha Rajeshwar:** Writing – review & editing, Validation. **Michelle K. Kidder:** Writing – review & editing, Supervision, Funding acquisition. **Jeremy C. Smith:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This material is based upon work supported by the U.S. Department of Energy, Office of Science, Biological and Environmental Research Program under award #ERKP752, and the DOE Office of Science, Office of Basic Energy Sciences, Division of Chemical Sciences, Geosciences, and Biosciences (CSGB) (Award No. DE-SC0022214; FWP 3ERKCG25). This manuscript has been authored by UT-Battelle, LLC, under contract DE-AC05-00OR22725 with the US Department of Energy (DOE). The US government retains and the publisher, by accepting the article for publication, acknowledges that the US government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this manuscript, or allow others to do so, for US government purposes. DOE will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (http://energy.gov/downloads/doe-public-access-plan).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.gce.2024.11.003.

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